

Joshua Sternadel

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jsternadel.github.io/portfolio/

Senior Software Engineer

Systems & Platform Engineering | Distributed Systems | Enterprise Automation

Software engineer with 15+ years of experience building enterprise systems across application development, infrastructure automation, distributed systems, data platforms, and full-stack engineering. Experienced taking complicated operational problems from architecture through implementation, deployment, and production support. Strongest at building the software and automation that connects systems that were never designed to work together.

U.S. Patent #10,366,069 — Systems and Methods for Database Management and Administration (Co-Author)

Engineering Highlights

- **Enterprise Data Migration:** Migrated 10+ years of Sprint Jira Data Center metadata and device certification data to Jira Cloud in 3 months with zero downtime for 3,000+ enterprise users, using a custom PHP hydration engine to unify remote JSON and local ORM data.
- **Distributed Systems:** Architected Porcupine, a RabbitMQ-based event bus connecting Jira workflows with downstream enterprise systems and automating workflow-driven notifications.
- **Data Abstraction:** Built a reflection- and annotation-driven PHP framework that unified SQL, NoSQL, JSON, and XML behind a single entity manager, reducing application boilerplate by 10x.
- **Infrastructure Automation:** Built an idempotent Ansible patch-management framework that automated remediation of a vulnerability backlog spanning 100+ monthly exploits across 97 hosts.
- **Enterprise Workflow Automation:** Replaced a decade-old manual SKU bottleneck with a human-in-the-loop workflow and full web application, then integrated the system with Porcupine for multi-stage TPM and reviewer approvals.
- **Infrastructure Reliability:** Replaced failing legacy infrastructure with a redundant VMware/iSCSI architecture and reduced enterprise downtime by 90% across three distributed locations.

- **Mobile Systems:** Developed an Android signal-smoothing algorithm using spherical geometry, acceleration, and last-known-speed data to reduce GPS coordinate jitter in lossy rail-yard and intermodal trucking environments.

Professional Experience

T-Mobile | Senior Software Engineer | 2021–2026

- Migrated 10+ years of Sprint Jira Data Center metadata and device certification data to Jira Cloud in 3 months with zero downtime for 3,000+ enterprise users, using a data-agnostic PHP framework originally developed at Sprint to hydrate and transform data across remote JSON APIs and local ORM sources.
- Built Node.js SDKs and Node-RED integrations with Ansible to capture playbook failures and automatically generate structured Jira incidents.
- Automated fleet mobilization across **132 Spirent Landslide traffic-generation VMs** using Ansible, programmatically updating Netplan configurations and staging changes to prevent host lockouts.
- Built an idempotent Ansible patch-management framework to automate remediation of a vulnerability backlog spanning **100+ monthly exploits across 97 hosts**.
- Delivered a serverless Angular Universal and Express-based 5G Speed Calculator using AWS Lambda microservices, eliminating dedicated application infrastructure and simplifying deployment.
- Architected the Branded Experience Portal as an Angular monorepo with reusable component libraries structured for future distribution as independent corporate NPM packages.
- Served as a technical advisor for high-risk infrastructure changes and enterprise deployments during the Sprint/T-Mobile integration.

Sprint | Application Developer III | 2017–2021

- Architected **Porcupine**, a RabbitMQ-based event bus that connected Jira workflows with downstream enterprise systems and automated workflow-driven notifications.
- Designed and built the frontend and workflow engine for a human-in-the-loop SKU automation platform that eliminated a decade-old enterprise bottleneck within 6 months.
- Integrated the SKU Portal with Porcupine to support multi-stage TPM and reviewer approval workflows.
- Architected and built a data-agnostic PHP framework using reflection, annotations, and a custom hydration engine to unify SQL, NoSQL, JSON, and XML behind a single entity-managed ORM.
- Reduced application boilerplate by approximately **10x** through reusable entity, persistence, serialization, and data-access abstractions.
- Validated a Dremio lakehouse architecture across MS SQL, MySQL, and MongoDB, demonstrating zero-ETL query federation across relational and NoSQL data sources.
- Built an observability platform across **45 VMs** using Docker consumers, .NET Core heartbeat telemetry, ELK, Prometheus, Grafana, and Rocket.Chat integrations.

- Served as a principal technical advisor for infrastructure and application engineering, including high-risk deployment reviews and infrastructure planning.

Sprint | Product Development Engineer II | 2013–2017

- Built a drag-and-drop prototyping platform for Out-of-Box Experience (OOBE) simulation, including validation engines that captured CPU, RAM, and battery telemetry from mobile devices.
- Hardened the Device Development and Certification requirements database for production by introducing component-based jQuery architecture, strict database transactions, and Jira-driven publishing controls.
- Built the device certification Excel import/export workflow into a controlled data exchange: generated constrained workbooks with only approved fields editable, validated and parsed returned files, and applied certification updates to the database within strict transactions that could roll back on errors.
- Led weekly engineering reviews with retail testing leadership to troubleshoot system anomalies and coordinate SLATE release timelines.

EXPERIS (Contractor at Sprint) | Technical Support Rep II | 2012–2013

- Wrote and maintained XML execution scripts for the automated SLATE diagnostic framework used across Sprint's national retail footprint.
- Improved legacy device-performance tracking applications to support automated on-site device validation at retail service centers.

MO-KAN Container Services | System Administrator | 2011–2012

- Replaced failing HP G3 bare-metal systems running Windows Server 2003 with a redundant, high-availability VMware environment.
- Designed and deployed VMware infrastructure backed by iSCSI SAN storage and Cisco switching through a vendor partnership.
- Reduced enterprise downtime by **90% across three distributed locations** through infrastructure modernization and redundancy.
- Managed disaster recovery, failover automation, network security, centralized domain architecture, and production infrastructure operations.

Additional Engineering Experience

Trucking Central | Software Consultant

- Developed early Android GPS and telematics applications for real-time intermodal logistics tracking.
- Built location-processing logic for environments where vehicle positioning data was subject to significant GPS loss and coordinate jitter.

Innovative Solutions | Mobile Solutions Technician

- Deployed and optimized Nextel communication devices and mobile diagnostic systems for municipal first responders and commercial operations.
- Supported mobile hardware and software deployments in field environments.

Patent

U.S. Patent #10,366,069 — Systems and Methods for Database Management and Administration

Co-author of a patented database-management system incorporating automated administration techniques and a custom SQL natural-sorting algorithm.

Technical Environment

Software Architecture: Distributed Systems · Event-Driven Architecture · Platform Engineering · API Design · Enterprise Integration · Data Abstraction · Developer Tooling · Internal Platforms

Languages & Backend: PHP · Node.js · TypeScript · JavaScript · C# · .NET Core · REST APIs · Native Modules

Frontend: Angular · Angular Universal · Vue.js 2 · TypeScript · JavaScript · jQuery · Client-Side Architecture

Data & Messaging: PostgreSQL · MS SQL · MySQL · MongoDB · RabbitMQ · Dremio · JSON · XML · SQL

Infrastructure & Automation: Ansible · Ansible Automation Platform · Infrastructure as Code · CI/CD · Docker · AWS Lambda · VMware vSphere/ESXi · iSCSI · Linux · Netplan

Observability: ELK · Prometheus · Grafana · Rocket.Chat · Telemetry Systems

Development Practices: Zero-Dependency Design · Custom SDK Development · ORM Design · Reflection · Annotation-Driven Architecture · Automation · Production Troubleshooting

Community Engineering

Seek and Serve (501c3) | Technical Advisory Board Member | 2026–Present

- Provide technical architecture and systems guidance supporting regional community outreach and operational expansion.

33 Meals | Technical Advisor & Infrastructure Volunteer | 2026–Present

- Designed and implemented the digital architecture and logistics pipeline supporting regional food-insecurity operations.
- Established core system deployments, secure domain configurations, and compliance planning for distribution operations.
- Designed a data-driven delivery framework to support scalable humanitarian food distribution.

Education

Associate of Arts (AA), Graphic Design | Johnson County Community College